

One Joint Transform for the Complete Exponential Cramér–Lundberg Ruin Atlas

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Abstract

For the classical compound-Poisson surplus with exponential claims, we derive one closed joint transform of ruin time and deficit. It gives ultimate ruin on both sides of the safety-loading wall and proves conditional overshoot independence. We also obtain both finite conditional ruin-time means and prove their critical infinite-mean boundary from a square-root root cusp. An adjustment martingale closes the exact all-time supremum mixture. No periodic-orbit or arithmetic target conclusion is drawn.

1 Frozen process and joint penalty

Let N_t be Poisson with rate $\nu \geq 0$, let Y_i be independent $\text{Exp}(\beta)$ claims with $\beta > 0$, and fix $c > 0, u \geq 0$. The surplus, strict ruin time, and deficit are

$$U_t = u + ct - \sum_{i=1}^{N_t} Y_i, \quad \tau = \inf\{t \geq 0 : U_t < 0\}, \quad D = -U_\tau. \quad (1)$$

Thus $u = 0$ is not initially ruined. The Route-A dynamical owner is the killed process

$$X_t = U_t \quad (t < \tau), \quad X_t = \Delta \quad (t \geq \tau),$$

whereas the joint transform below is a first-passage functional of the underlying surplus U . Gerber and Shiu [1] are cited only for discounted-penalty lineage; Dreikic and Willmot [2] are cited only for exponential-claim ruin-time lineage. No formula or proof below is outsourced.

For $q, s \geq 0$ put

$$\Phi_{q,s}(u) = \mathbb{E}_u[e^{-q\tau - sD}; \tau < \infty]$$

and define

$$r_q = \frac{c\beta - \nu - q + \sqrt{(c\beta - \nu - q)^2 + 4c\beta q}}{2c}. \quad (2)$$

Theorem 1 (Complete joint transform). *For every frozen parameter,*

$$\boxed{\Phi_{q,s}(u) = \frac{\beta - r_q}{\beta + s} e^{-r_q u}.} \quad (3)$$

Here $0 \leq r_q \leq \beta$. If $\nu = 0$, then $r_q = \beta$ and (3) is zero. If $\nu > 0$, conditional on ruin, $D \sim \text{Exp}(\beta)$ and D is independent of τ .

Proof. First-jump conditioning gives, for $u > 0$,

$$\begin{aligned} 0 &= c\Phi'(u) - (\nu + q)\Phi(u) + \nu \int_0^u \Phi(u-y)\beta e^{-\beta y} dy \\ &\quad + \nu \int_u^\infty e^{-s(y-u)}\beta e^{-\beta y} dy. \end{aligned} \quad (4)$$

The last integral is $\beta e^{-\beta u}/(\beta + s)$. To prove that an exponential ansatz is exhaustive, introduce the convolution state

$$J(u) = \int_0^u \Phi(u-y) \beta e^{-\beta y} dy.$$

Equation (4) is equivalent to the two-dimensional inhomogeneous system

$$\begin{aligned} c\Phi' &= (\nu + q)\Phi - \nu J - \frac{\nu\beta}{\beta + s} e^{-\beta u}, \\ J' &= \beta(\Phi - J), \quad J(0) = 0. \end{aligned} \tag{5}$$

Its homogeneous mode e^{-ru} exists exactly when

$$cr^2 - (c\beta - \nu - q)r - q\beta = 0, \tag{6}$$

so these two modes exhaust the homogeneous solution space (with the usual generalized mode at a double root). A particular solution of (5) is $(\Phi, J) = (0, -\beta e^{-\beta u}/(\beta + s))$. For $q > 0$, the two roots have opposite signs, so boundedness removes the negative root and retains (2). At $q = 0$ with $0 < \nu < c\beta$, both 0 and $R = \beta - \nu/c$ give bounded nonnegative local solutions. Here the strong law for $Z_t = \sum_{i \leq N_t} Y_i - ct$ gives $Z_t/t \rightarrow \nu/\beta - c < 0$, hence $M = \sup_{t \geq 0} Z_t < \infty$ almost surely and $\Phi_{0,s}(u) \leq \mathbb{P}(M > u) \rightarrow 0$; this removes the constant mode and selects R . If $\nu > c\beta$, boundedness removes the growing negative-root mode and retains $r = 0$. At equality the second generalized mode grows linearly and is again removed by boundedness. For the surviving mode, $J = A\beta e^{-ru}/(\beta - r) - \beta e^{-\beta u}/(\beta + s)$, so $J(0) = 0$ forces $A = (\beta - r)/(\beta + s)$. This mode exhaustion plus the stated chamber boundaries proves uniqueness, rather than assuming it from the ansatz. If $\nu = 0$ there are no claims and $\Phi \equiv 0$, which is (3) with $r_q = \beta$. Continuity under the common-path coupling extends the solution from $u > 0$ to $u = 0$; equality at a claim has probability zero, so the strict-passage convention is preserved.

For $\nu > 0$, (3) factors as $\Phi_{q,s} = \{\beta/(\beta + s)\}\Phi_{q,0}$. This identifies the conditional Laplace transform of D and its independence from τ . Pathwise, the same fact is exponential memorylessness beyond the pre-ruin reserve. $\square$

2 Every safety-loading chamber

Put $\rho = \nu/(c\beta)$. Setting $q = s = 0$ in (2)–(3) gives the following complete boundary law.

Theorem 2 (Ruin probability and conditional mean time). *For $\nu > 0$:*

chamber	$\mathbb{P}_u(\tau < \infty)$	$\mathbb{E}_u[\tau \mid \tau < \infty]$
$\rho < 1$	ρe^{-Ru} , $R = \beta - \nu/c$	$\frac{1 + \nu u/c}{c\beta - \nu}$
$\rho = 1$	1	∞
$\rho > 1$	1	$\frac{\beta u + 1}{\nu - c\beta}$

(7)

In all three chambers, $\mathbb{E}[D \mid \tau < \infty] = 1/\beta$. If $\nu = 0$, ruin is impossible and conditional ruin quantities are undefined.

Proof. At $q = 0$, (2) is $R > 0$ when $c\beta > \nu$ and zero otherwise, giving the middle column. Differentiate $\Phi_{q,0}$ at zero discount. In the profitable chamber, $r'_0 = \nu/[c(c\beta - \nu)]$; in the adverse chamber, $r'_0 = \beta/(\nu - c\beta)$. Logarithmic differentiation gives the two finite means in (7).

At criticality,

$$r_q = \frac{-q + \sqrt{q^2 + 4c\beta q}}{2c} \sim \sqrt{\frac{\beta q}{c}}.$$

Hence the right derivative of $\Phi_{q,0}$ at zero is $-\infty$, proving the infinite mean rather than substituting into a singular side formula. The deficit mean follows from the first theorem. $\square$

3 Adjustment martingale and exact supremum

Assume $0 < \rho < 1$ and set $Z_t = \sum_{i=1}^{N_t} Y_i - ct$. Since

$$\nu \left(\frac{\beta}{\beta - R} - 1 \right) - cR = 0, \quad R = \beta - \nu/c, \quad (8)$$

e^{RZ_t} is a mean-one martingale. If $M = \sup_{t \geq 0} Z_t$, ruin from reserve u is precisely $\{M > u\}$. Therefore

$$\mathbb{P}(M = 0) = 1 - \rho, \quad \mathbb{P}(M > u) = \rho e^{-Ru} \quad (u \geq 0). \quad (9)$$

Thus the positive part has density $\rho R e^{-Rx}$. Equation (9) is a source-local workload duality, not a new queue spectrum or target bridge.

4 Evidence and Route-A boundary

The canonical receipt contains 36 exact regime rows, 448 joint-transform rows, 144 conditional-first-mean rows, 12 martingale/supremum rows, and six boundary rows. An independent checker passes 4,487 assertions; SymPy passes 15 identities; fresh replay is byte-exact; and 26/26 repaired-hash or stale-hash mutations are rejected. Finite rows are regression checks, not proof of the continuum theorem.

Release item	Result
Regime / transform / first mean / martingale / boundary	36/448/144/12/6
Independent assertions / symbolic identities	4,487/15 PASS
Fresh replay / hostile mutations	PASS / 26/26 rejected
Evidence SHA-256	6551879f2a73...ded29f0

The nearest finite queue owner studies a birth–death spectrum; it does not contain this continuous-reserve, absorbing, joint deficit/time theorem. This workspace distinction is not a literature-priority claim.

The Route-A tuple is

$$(A0_FAIL, A1_FAIL, A2_FAIL, A3_FAIL, A4_FAIL), \quad \text{ROUTE_A_REJECTED}. \quad (10)$$

There is no arithmetic origin. More precisely, the killed PDMP/Markov semigroup has no intrinsic deterministic, enumerable primitive-periodic-orbit owner. No rational-prime carrier, logarithmic clock, target determinant, target zero match, Hilbert–Pólya operator, or Route-B input follows. Scope is NO_BAD_EULER_OR_ROOT_NUMBER.

References

- [1] H. U. Gerber and E. S. W. Shiu, “On the time value of ruin,” *North American Actuarial Journal* **2** (1), 48–72 (1998), doi:10.1080/10920277.1998.10595671.
- [2] S. Drekić and G. E. Willmot, “On the density and moments of the time of ruin with exponential claims,” *ASTIN Bulletin* **33** (1), 11–21 (2003), doi:10.2143/AST.33.1.1036.